

University Database Schema

- **classroom**(*building*, room number, capacity)
- **department**(*dept name*, building, budget)
- **course**(*course id*, title, dept name, credits)
- **instructor**(*ID*, name, dept name, salary)
- **section**(*course id*, sec id, semester, year, building, room number, time slot id)
- **teaches**(*ID*, course id, sec id, semester, year)
- **student**(*ID*, name, dept name, tot cred)
- **takes**(*ID*, course id, sec id, semester, year, grade)
- **advisor**(*s ID*, *i ID*)
- **time slot**(*time slot id*, day, start time, end time)
- **prereq**(*course id*, prereq id)

Employee Database Schema

- **employee**(employee_name, street, city)
- **works**(employee_name, company_name, salary)
- **company**(company_name, city)
- **manages**(employee_name, manager_name)

SQL Queries and Questions

Question 1

For all students in the university who have taken some course, find their names and the course ID of all courses they took.

```
SELECT name, course_id
FROM student NATURAL JOIN takes;
```

```
SELECT name, course_id
FROM student, takes
WHERE student.ID = takes.ID;
```

```
SELECT name, course_id
FROM student JOIN takes USING (ID);
```

```
SELECT name, course_id
FROM student JOIN takes ON student.ID = takes.ID;
```

Question 2

List the names of students along with the titles of courses that they have taken.

```
SELECT name, title
FROM student NATURAL JOIN takes NATURAL JOIN course;
```

```
SELECT name, title
FROM student NATURAL JOIN takes, course
WHERE takes.course_id = course.course_id;
```

```
SELECT name, title
FROM (student NATURAL JOIN takes) JOIN course USING (course id);
```

Question 3

Find all students who have not taken a course.

```
SELECT ID
FROM student NATURAL LEFT OUTER JOIN takes
WHERE course_id IS NULL;
```

Question 4

Find all students from the Computer Science department along with courses they took in Spring 2017, including those who did not take any course in that semester.

```
SELECT *
FROM
  (SELECT * FROM student WHERE dept_name = 'Comp. Sci.') AS s
NATURAL FULL OUTER JOIN
  (SELECT * FROM takes WHERE semester = 'Spring' AND year = 2017) AS t;
```

Question 5

Find a list of titles of all courses taught in Spring 2017 along with the name of the instructor.

```
SELECT course.title, instructor.name
FROM course
NATURAL JOIN section
NATURAL JOIN teaches
NATURAL JOIN instructor
WHERE semester = 'Spring' AND year = 2017;
```

```
SELECT course.title, instructor.name
FROM course
JOIN section
    ON course.course_id = section.course_id
JOIN teaches
    ON section.course_id = teaches.course_id
    AND section.sec_id = teaches.sec_id
    AND section.semester = teaches.semester
    AND section.year = teaches.year
JOIN instructor
    ON teaches.ID = instructor.ID
WHERE section.semester = 'Spring'
    AND section.year = 2017;
```

Question 6

Display a list of all instructors, showing each instructor's ID and the number of sections taught. Make sure to show the number of sections as 0 for instructors who have not taught any section. Use an outer join, and do not use subqueries.

```
SELECT i.ID, COUNT(s.course_id) AS section_count
FROM instructor AS i
LEFT JOIN teaches AS s
    ON i.ID = s.ID
GROUP BY i.ID;
```

Question 7

Rewrite the query from part (6) using a scalar subquery, without using outer joins.

```
SELECT i.ID,
    (SELECT COUNT(*)
     FROM teaches AS s
     WHERE s.ID = i.ID) AS section_count
FROM instructor AS i;
```

Question 8

Display the list of all course sections offered in Spring 2018, along with the ID and name of each instructor teaching the section. If a section has multiple instructors, it should appear multiple times in the result. If a section does not have any instructor, it should appear with the instructor name set to '—'.

```

SELECT s.course_id, s.sec_id, s.semester, s.year,
       COALESCE(i.ID, '-') AS instructor_id,
       COALESCE(i.name, '-') AS instructor_name
FROM section AS s
LEFT JOIN teaches AS t
      ON s.course_id = t.course_id
      AND s.sec_id = t.sec_id
      AND s.semester = t.semester
      AND s.year = t.year
LEFT JOIN instructor AS i
      ON t.ID = i.ID
WHERE s.semester = 'Spring'
      AND s.year = 2018;

```

Question 9

Display the list of all departments, with the total number of instructors in each department, without using subqueries. Make sure to show departments that have no instructors, listing them with an instructor count of zero.

```

SELECT d.dept_name, COUNT(i.ID) AS instructor_count
FROM department AS d
LEFT JOIN instructor AS i
      ON d.dept_name = i.dept_name
GROUP BY d.dept_name;

```

Question 10

For the employee database, find those employees with no manager. Write the query first using an outer join, and then again without using an outer join.

(a) Using Outer Join

```

SELECT employee_name
FROM employee
LEFT JOIN manages
      ON employee.employee_name = manages.employee_name
WHERE manager_name IS NULL;

```

(b) Without Outer Join

```

SELECT e.employee_name
FROM employee AS e
WHERE NOT EXISTS (
  SELECT 1
  FROM manages AS m
  WHERE e.employee_name = m.employee_name
);

```

Question 11

Define a view `tot_credits(year, tot_credits)` that gives the total number of credits taken by students in each year.

```
CREATE VIEW tot_credits (year, tot_credits) AS
SELECT year, SUM(credits)
FROM takes
NATURAL JOIN course
GROUP BY year;
```

Note: This solution assumes that there is no year in which students did not take any course, even if sections were offered in that year.